Dr. Weymar Andrés Astaiza Sulez

Professor of Mathematics
Universidad del Cauca, Colombia

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1 Short Bio

Weymar Andrés Astaiza Sulez is a mathematician and researcher currently serving as an adjunct Professor at the Universidad del Cauca, Colombia. He received his Ph.D. in Applied Mathematics from the University of São Paulo, Brazil, in 2024, and his Master's degree in Mathematics from Universidad de Los Andes, Colombia. His research interests include partial differential equations, semigroup theory, pseudodifferential operators, functional analysis, and computational mathematics. He has published research articles in mathematical analysis and is currently working on pseudodifferential operators and related problems. He is also involved in the organization of mathematical activities through UMALCA–ICMAM and serves as an associate researcher in the ICMAM Latin America Research Group in Analysis and PDE.

Research Interests

Keywords: Partial differential equations; pseudodifferential operators; functional analysis; analytic semigroups; operator theory; applied mathematics; computational mathematics.

Academic Positions

Adjunct Instructor – Mathematics
Universidad del Cauca, Colombia March 2023–Present

- Teaching Computational Algorithms and undergraduate Mathematics courses.
- Research in Partial Differential Equations, Pseudodifferential Operators, Functional Analysis, and related areas of Analysis.

Online Assistant Professor
Universidade Virtual do Estado de São Paulo (UNIVESP), Brazil October 2022–January 2023

Adjunct Instructor
Universidad de Los Andes, Colombia August 2017–May 2018

Adjunct Instructor
SENA Regional Cauca, Colombia April–December 2016

Adjunct Instructor
SENA Regional Valle, Colombia February–April 2016

2 Education

Ph.D. in Applied Mathematics February 2024
University of São Paulo, Brazil
Dissertation: *Parabolic Equations on Conic Manifolds*
Advisor: Prof. Pedro Tavares Paes Lopes

M.Sc. in Mathematics July 2017
Universidad de Los Andes, Bogotá, Colombia
Degree Project: *The Monte Carlo Method and Its Applications to Elliptic Equations*
Advisor: Prof. Jean Carlos Cortissoz Iriarte

B.Sc. in Mathematics September 2013
Universidad del Cauca, Colombia
Degree Project: *The Babuška–Brezzi Theory and Its Applications to the Stokes and Darcy Equations*
Advisor: Prof. Ramiro Acevedo

Scholarships and Funding

CAPES Doctoral Scholarship
Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES), Brazil.

Doctoral studies in Applied Mathematics at the University of São Paulo.

Graduate Scholarship
Universidad de los Andes, Colombia.

Scholarship supporting Master's studies in Mathematics.

Research Experience

Doctoral Research – Partial Differential Equations
University of São Paulo, Brazil 2019–2024

Research on parabolic partial differential equations on conic manifolds, with emphasis on functional analytic methods and operator theory.

Master's Research – Numerical Methods for PDEs
Universidad de los Andes, Colombia 2015–2017

Research on Monte Carlo methods and their applications to elliptic partial differential equations.

Undergraduate Research – Numerical Analysis and PDEs
Universidad del Cauca, Colombia 2012–2013

Research on Babuška–Brezzi theory and its applications to the Stokes and Darcy problems.

Collaborative Research – Graph Theory

Research on symmetric tensor powers of graphs, resulting in a publication in *Scientia Series A: Mathematical Sciences*.

Publications

1. Astaiza Sulez, W. A., Barrios, A. J., Chimal-Dzul, H., Garcia, S. R., Lopez de la Luz, J., Moll, V. H., Puig, Y., & Villamizar, D.

   Symmetric Tensor Powers of Graphs.

   Scientia Series A: Mathematical Sciences, Vol. 36 (2026), pp. 13–35.

2. Astaiza Sulez, W. A. & Lopes, P. T. P.

   Sectorial Pseudodifferential Operators in $L_1(\mathbb{R}^n)$ and $BUC(\mathbb{R}^n)$.

   In *Analysis and PDE in Latin America (ICMAM 2024)*, Trends in Mathematics, Vol. 15, pp. 25–31. Springer, 2026.

   First published online: January 2, 2026.

3. Book: Quintero, J. R., Delgado, J., Gallego, F. A., Astaiza Sulez, W. A., Cardona Sánchez, D., eds.

   Analysis and PDE in Latin America: ICMAM 2026.

   Trends in Mathematics, Springer Birkhäuser.

   Forthcoming, 2027.

Manuscripts in Preparation

1. Astaiza Sulez, W. A. & Lopes, P. T. P.

   Manuscript on Partial Differential Equations.

   Manuscript in preparation; planned for submission in 2026.

Conference Presentations

1. Astaiza Sulez, W. A.

   Sectorial Pseudodifferential Operators on $L_1(\mathbb{R}^n)$ and $BUC(\mathbb{R}^n)$.

   XXIV Congreso Colombiano de Matemáticas, Sociedad Colombiana de Matemáticas, Universidad Tecnológica de Pereira, Pereira, Colombia, June 2–6, 2025.

   Oral Presentation.

Teaching Experience

Adjunct Instructor – Computational Algorithms and Mathematics
Universidad del Cauca, Colombia March 2023–Present

- Teaching Computational Algorithms and undergraduate Mathematics courses.
- Development of lectures, computational exercises, programming assignments, and mathematical applications.
- Use of Python and computational tools for mathematical problem solving and algorithmic development.

Adjunct Instructor – Mathematics
Fundación Universitaria de Popayán, Colombia February 2024–June 2026

- Teaching undergraduate Mathematics courses.
- Preparation of lectures, exercises, assessments, and mathematical applications.

Online Assistant Professor
Universidade Virtual do Estado de São Paulo (UNIVESP), Brazil October 2022–January 2023

- Online teaching and academic support for undergraduate Mathematics courses.
- Assistance with course activities, student questions, and mathematical content.

Teaching Monitor
University of São Paulo, Brazil March–July 2022

Teaching Monitor
University of São Paulo, Brazil March–July 2020

Adjunct Instructor – Undergraduate Mathematics
Universidad de Los Andes, Colombia August–November 2017

Adjunct Instructor – Undergraduate Mathematics
Universidad de Los Andes, Colombia January–May 2018

Teaching Assistant
Universidad de Los Andes, Colombia January–May 2014

Teaching Assistant
Universidad de Los Andes, Colombia August–November 2014

Teaching Assistant
Universidad de Los Andes, Colombia January–May 2015

Teaching Assistant
Universidad de Los Andes, Colombia January–May 2017

Adjunct Instructor
SENA Regional Cauca, Colombia April–December 2016

- Teaching Mathematics and related quantitative subjects.

Adjunct Instructor
SENA Regional Valle, Colombia February–April 2016

- Teaching Mathematics and related quantitative subjects.

Graduate and Research Assistantships

Ph.D. Graduate Assistant
University of São Paulo, Brazil June 2021–November 2022

Research Assistant
Universidad de Los Andes, Colombia June–July 2015

3 Academic Service

Organizing Committee Member
UMALCA–ICMAM Latin American Mathematics Seminar 2025–Present

- Member of the organizing committee, contributing to the organization of scientific activities and academic events.

Associate Researcher
ICMAM Latin America – Research Group in Analysis and PDE 2025–Present

- Research sponsor: Prof. Duván Cardona Sánchez, King Fahd University of Petroleum and Minerals, College of Computing and Mathematics, Dhahran, Saudi Arabia.

Editorial Activities

Editor
Analysis and PDE in Latin America: ICMAM 2026
Springer / Birkhäuser 2026–Present

4 Languages

- Spanish: Native
- Portuguese: Fluent
- English: Proficient

References

Prof. Pedro Tavares Paes Lopes
Associate Professor
University of São Paulo, São Paulo, Brazil
Email: pplopes@icmc.usp.br | pplopes@ime.usp.br
Ph.D. Advisor

Prof. Jean Carlos Cortissoz Iriarte
Associate Professor
Universidad de los Andes, Bogotá, Colombia
Email: jcortiss@uniandes.edu.co
M.Sc. Advisor

Prof. René Joaquín Meziat Vélez
Associate Professor
Universidad de los Andes, Bogotá, Colombia
Email: rmeziat@uniandes.edu.co
Former Teaching Assistant Supervisor

Prof. Duván Cardona Sánchez
Assistant Professor of Mathematics

King Fahd University of Petroleum and Minerals
College of Computing and Mathematics
Dhahran, Kingdom of Saudi Arabia
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